ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

FACT CHECK · C3 · ATOMS AND CONSERVATION

Atoms and mass

★☆☆
Grade 5

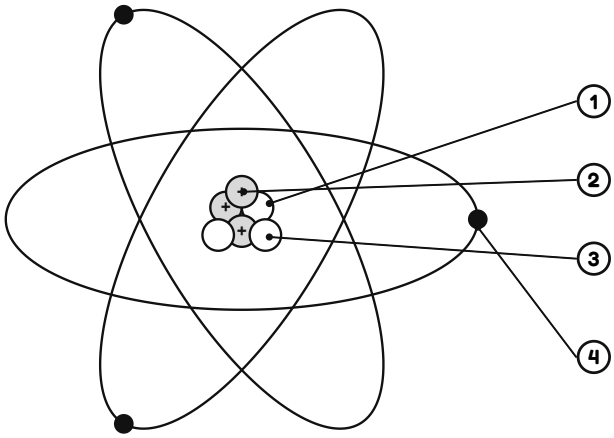
Learning goals

- Label an atom.
- Recall the law of conservation of mass.

1 **Label the atom.** [4]

LABELS

neutron **nucleus** **electron** **proton**



(1) nucleus (2) proton

(3) neutron (4) electron

2 **Fill in the gaps.** [5]

WORD BANK

element **bigger** **closed** **destroyed** **open** **mass** **atoms**

All matter is made of tiny atoms. A substance made of only one kind of atom is an element. In physical and chemical changes, atoms are not made or destroyed, so the total mass stays the same. A sealed bag is a closed system.

3 **Work out the missing masses.** [3]

Change	Mass before	Mass after
ice melting	100 g of ice	100 g of water
dissolving sugar	200 g water + 15 g sugar	215 g of sugar solution
freezing juice	330 g of juice	330 g of ice lolly

4 **True or false?** [4]

Statement	True	False	If false, correct it
1. When sugar dissolves, its atoms are destroyed.		✔	The atoms are still there, spread through the water.
2. Water is made of hydrogen and oxygen atoms.	✔		
3. In a nuclear change, one element can turn into another.	✔		
4. Burning wood makes the atoms disappear.		✔	The atoms are rearranged into gases, smoke and ash.

ANSWER KEY

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THINK IT THROUGH · C3 · ATOMS AND CONSERVATION

Where did the mass go?

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Grade 5

Learning goals

- Explain changes in mass using open and closed systems.
- Apply conservation to new examples.

1 **A candle has a mass of 50 g. After burning for an hour its mass is 42 g. Has mass been destroyed? Explain.** [2]

No: the wax reacted with oxygen and turned into carbon dioxide gas and water vapour, which escaped into the air (an open system).

2 **Describe an experiment to show that mass is conserved when a candle burns.** [2]

Burn it inside a sealed (closed) container on a balance; the total mass before and after would be the same (with an adult, safely).

3 **An iron nail gets heavier as it rusts. Where does the extra mass come from?** [2]

Iron atoms join with oxygen (and water) from the air; the added atoms add mass.

4 **Use the idea of conservation to check this nuclear change. Are the protons and neutrons conserved?** [2]

Before	After
a nucleus with 235 protons + neutrons, hit by 1 neutron	139 + 94 + 3 free neutrons

Before: 235 + 1 = 236. After: 139 + 94 + 3 = 236. Yes, the total is the same.

ANSWER KEY

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INVESTIGATE · C3 · ATOMS AND CONSERVATION

Is mass conserved?

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Grade 5

Learning goals

- Measure mass accurately.
- Compare open and closed systems.
- Explain unexpected results.

You will need

a digital balance, a zip-lock bag, a small cup, vinegar, baking soda, a teaspoon, an open plastic cup

Stay safe

Wear goggles. Vinegar can sting your eyes. Point the bag away from faces.

Class 5C carried out three tests and measured the mass before and after.

1 **Complete the table.** [3]

Test	Mass before (g)	Mass after (g)	Change (g)
dissolving salt in water (open cup)	270	270	0
vinegar + baking soda in a sealed bag	312	312	0
vinegar + baking soda in an open cup	312	309	−3

2 **Why did the mass go down in the open cup but not in the sealed bag?** [2]

The reaction made a gas (carbon dioxide). In the open cup the gas escaped into the air; in the sealed bag it was trapped, so all the atoms stayed.

3 **What evidence showed that a gas was made?** [1]

Bubbles and fizzing; the sealed bag puffed up.

4 **Write a conclusion about conservation of mass.** [2]

In a closed system mass is conserved: no atoms are gained or lost, even in a chemical change.

5 **The balance shows whole grams only. How could this affect the results?** [1]

Small changes of less than 1 g would not show; a more precise balance (0.1 g) would be better.

ANSWER KEY

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REAL-WORLD CASE STUDY · C3 · ATOMS AND CONSERVATION

Energy from atoms: Rooppur

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Grade 5

Learning goals

- Explain nuclear power simply.
- Weigh advantages and disadvantages.
- Consider different perspectives.

Science · Geography · Economics · Ethics · Environment

Bangladesh needs more and more electricity for homes, schools, hospitals and factories. Most of it comes from burning **natural gas**, which is running out and gives off carbon dioxide.

At **Rooppur**, in Pabna district on the bank of the Padma, Bangladesh is building its **first nuclear power plant**, with help from Russia, and it is starting up in the mid-2020s. Inside its reactors, uranium atoms are split. The heat boils water into steam, which spins turbines to make electricity. A small amount of uranium can produce a very large amount of energy, and the plant gives off almost no carbon dioxide while working.

But nuclear power also has challenges: it is very **expensive** to build; it must be run with great **safety** care; and the used fuel stays **radioactive** (dangerous) for thousands of years, so it must be stored very carefully.

Based on public information about the Rooppur Nuclear Power Plant.

1 **Put the steps of making electricity at Rooppur in order.** [3]

4 Steam spins a turbine.

1 Uranium atoms split in the reactor.

5 The turbine turns a generator, making electricity.

3 The heat boils water into steam.

2 This releases a lot of heat.

2 **Complete the table.** [6]

Advantages of nuclear power	Disadvantages of nuclear power
a lot of energy from a little fuel	very expensive to build
almost no carbon dioxide while working	used fuel is radioactive for thousands of years
steady power day and night	accidents, though rare, can be very serious

3 **What might each person think about the plant?** [3]

Person	View
a factory owner in Dhaka	More reliable electricity will help my business grow.
a farmer living near Rooppur	Will my land and the river stay safe? Will there be jobs for us?
a climate scientist	It cuts carbon dioxide compared with gas, but solar and wind are also needed.

4 **Your opinion: should Bangladesh get more electricity from nuclear power, solar power or both? Give reasons.** [2]

Accept balanced reasoning, e.g. both: nuclear for steady power, solar for clean cheap power in the day; solar needs land and storage.

ANSWER KEY

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REVIEW AND CHALLENGE · C3 · ATOMS AND CONSERVATION

Show what you know

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Grade 5

Learning goals

- Use key vocabulary.
- Answer test-style questions.

1 **Find the key words.**

A	T	O	M	V	S	V	T	H	J	Y
D	A	W	S	T	N	P	I	C	E	G
B	O	R	E	F	M	A	S	S	G	P
T	F	M	G	L	P	R	O	T	O	N
K	S	E	A	L	E	D	T	C	K	E
I	F	O	S	Y	T	M	Y	F	U	M
O	E	Y	N	U	C	L	E	U	S	D
E	L	E	C	T	R	O	N	N	L	L
R	H	K	M	J	S	I	A	W	T	A
C	M	U	M	O	L	E	C	U	L	E
Y	N	E	U	T	R	O	N	W	A	I

atom **electron** **element** **mass** **molecule** **neutron** **nucleus** **proton** **sealed**

2 **Salt (20 g) is dissolved in water (180 g). What is the mass of the solution?** [1]

- ○ 160 g
- ○ 180 g
- ● **200 g**
- ○ 220 g

3 **Which is *a* closed system?** [1]

- ○ an open pan of boiling water
- ● ***a* sealed jar**
- ○ *a* candle burning in *a* room
- ○ *a* puddle drying

4 **Draw particle pictures to show that 10 atoms before *a* change are still 10 atoms after.** [2]

before

10 atoms drawn, some joined in pairs.

after

The same 10 atoms, rearranged into different groups.

5 **Challenge. Explain why scientists say atoms are conserved in chemical changes but not in nuclear changes, and what is still conserved in a nuclear change.** [3]

In chemical changes atoms only rearrange, so the same atoms are there before and after. In nuclear changes the nucleus changes, so atoms of one element become another; but the total number of protons plus neutrons stays the same.